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LETTER

Typology of U.S. transit systems reveals energy data gap and potential for diesel-based emissions reductions

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Supplementary material for this article is available online

Abstract

We present a typology of 2205 U.S. transit systems, derived from 167 operational indicators in the National Transit Database, to reveal distinct patterns. Using kernel principal component analysis and Gaussian mixture modeling, we identify five distinct transit system types: *Metro Bus–Rail*, *Mid-Sized Urban*, *Hybrid Multimodal*, *Urban Specialized*, and *Diverse Operations*. We also discover a gap in diesel consumption reporting (76% of systems), which are predominantly of the *Mid-Sized Urban* type. To bridge this critical data gap, we apply extreme gradient boosting to predict the diesel consumption for 1674 underreported systems. We then estimate lifecycle emissions, accounting for regional grid carbon intensity. The result highlights significant variability in diesel consumption and emissions across transit types, with *Metro Bus–Rail* systems contributing the highest emissions. To evaluate decarbonization potential, we simulate fleet electrification and grid decarbonization. Our scenario analyses indicate that up to 82% of 10.27 $MMTCO_2e$ diesel-based transit emissions can be eliminated through fleet electrification, with further grid decarbonization eliminating the remainder. These findings provide actionable insights for policymakers to prioritize electrification, optimize resource allocation, and support sustainable transit strategies in the U.S.

1. Introduction

Transportation is a major contributor to greenhouse gas (GHG) emissions, accounting for 28% of total energy use and 67% of total petroleum consumption in the U.S. in 2021 [1]. At the national policy level, modernizing and expanding public transit has been identified as a key pillar alongside vehicle electrification and fuel economy improvements [2]. Public transit systems provide essential mobility, reduce congestion, and improve air quality, offering significant opportunities to cut emissions through decarbonization, increased ridership, and reduction in private car usage [3, 4]. Efforts to decarbonize public transit are accelerating under federal initiatives such as the U.S. low- or no-emission vehicle program, which provides funding to state and local governmental authorities [5]. Such transitions can deliver not only climate benefits but also public health improvements through reduced air pollution exposure [6]. In 2022, there were 2253 transit systems in operation across the U.S., including 1308 rural and 945 urban systems [7]. These systems differ in terms of geography, network topology, and population served. While buses remain the dominant mode, many agencies, particularly in rural areas, rely heavily on demand-responsive services. This diversity makes it difficult to generalize policies or investments. Thus, an understanding of key underlying patterns in structure, operation, supply, demand, and energy is critical to support further nationwide transit decarbonization initiatives. Furthermore, without comprehensive system-level energy and emissions data, agencies face the challenge of identifying priorities and tracking their progress toward reaching national climate goals.

Typology analysis groups systems with similar characteristics to reveal structural patterns that support benchmarking and policy design. Early work

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explored form-function typologies through empirical clustering of city characteristics [8, 9]. Later studies examined physical network characteristics [10, 11], as well as the spatial structure of mobility networks [12]. Other studies expanded to geospatial and spatiotemporal typologies of transportation systems, classifying systems or travel patterns based on spatial structure [13], built-environment form [14], or geographic-temporal passenger flow patterns [15]. Beyond system-wide classifications, typology frameworks have been applied to specific travel behaviors, such as cyclist profiles [16] and electric vehicle user types with charging session patterns [17], as well as roadway safety [18]. Typology approaches have also been widely applied in related areas such as transit-oriented development planning, where clustering methods classify neighborhoods or station areas by built-environment characteristics and access levels [19, 20]. Recent work has developed multidimensional typologies within large metropolitan regions by integrating trip demand, multimodal networks, traffic flow, and land use to identify peer groups with shared transportation dynamics [21]. Other research has created emissions-focused typologies linking population density, climate, mode shares, street network characteristics, and built-up area to differences in per-capita GHG emissions across metropolitan areas [22].

Few typology studies have examined transit systems directly, especially at the national scale, where the diversity of operating contexts and service types requires more refined clustering. Earlier efforts clustered a limited number of transit systems (typically 120–250) based on supply-side indicators such as service miles, fleet size, and operating expenses [23–25]. These studies demonstrated that grouping agencies by operating characteristics can improve performance benchmarking and policy evaluation. More recent approaches have expanded the framework to include demographic and structural factors correlated with ridership [26], but often excluded rural systems and overlooked operational dimensions. Rural and small urban agencies face distinct demographic, financial, and governance challenges that affect service provision and sustainability [27]. However, detailed reporting on rural and small urban transit systems did not begin until 2009, largely due to data collection costs [28]. Only two efforts have since analyzed these systems using clustering methods, each constrained by a small number of variables [29, 30]. Yet, no typology study has examined transit systems at the national scale while incorporating a comprehensive set of variables, spanning demographic, geographic, operational, financial, and service characteristics, including mode-specific measures such as fleet composition, energy use, and ridership patterns.

Energy and emissions tracking is key to sustainable decarbonization of transit systems. A few studies have developed vehicle- and system-level models that integrate multiple bus powertrains into comprehensive frameworks, but these have typically focused on specific transit systems [31–33] or regions [34–36]. However, there is still a need for a broader framework that can estimate transit system emissions and decarbonization potential on a national scale. Importantly, operational emissions represent only part of the total climate impact. Life-cycle contributions from vehicle manufacturing, infrastructure construction, and fuel production can be substantial [37]. The emissions benefits of electrification also depend heavily on the carbon intensity of the regional power grid, with full reductions realized only as electricity generation decarbonizes [38].

To address these gaps, this study makes several key contributions. First, we develop a novel typology of U.S. transit systems, both rural and urban, into operationally and contextually similar groups using 167 indicators. To our knowledge, this is the first typology accounting for every single transit system in the U.S. It captures the diversity of service types and operating environments, enabling comparisons and the transfer of insights from data-rich systems to those with limited reporting. Our results reveal that 76% of transit systems do not report energy data. Moreover, of the five types uncovered, the most frequently occurring accounts for all the unreported energy data. This makes it difficult for decision makers and transit authorities to identify priorities and track progress toward national climate goals. Thus, we fit a predictive model to estimate diesel consumption and associated emissions for the 1674 underreported systems. The model addresses a significant national data gap by generating system-level emissions estimates for the entire U.S. transit network.

2. Method

We developed a three-stage framework to analyze the typology of U.S. transit systems and estimate their diesel emissions (figure 1). First, we reduced the dimensionality of the dataset using kernel principal components analysis (kPCA). This step was crucial in handling the high-dimensional data and retaining the most significant features. Second, we applied a Gaussian mixture model (GMM) to the reduced dataset to cluster the transit agencies into distinct types. This typology enabled us to identify patterns and similarities among different transit systems. Third, we estimated a predictive model for diesel consumption using extreme gradient boosting (XGBoost), trained on systems with available National Transit Database (NTD) energy data and

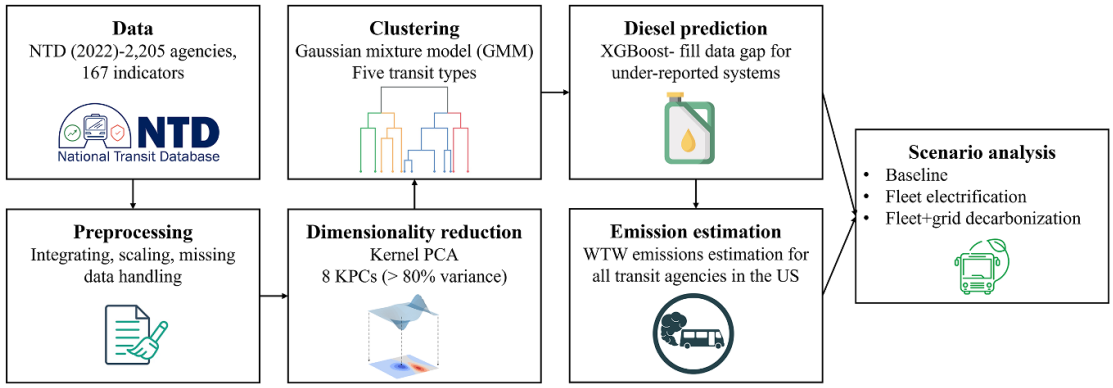


Figure 1. Framework for transit typology, diesel prediction, and emissions estimation.

incorporating the identified typology as an explanatory feature. This model allowed us to predict diesel emissions for agencies without reported energy data, quantifying the national emissions gap and the potential for reductions.

2.1. Data

We obtained the data used in this study from the NTD, which is publicly available from the Federal Transit Administration [28]. The NTD contains multiple datasets (including operations, performance, demand, and supply) on every transit agency operating in the U.S. We selected data for the year 2022 (table 1). We extracted 190 variables from the NTD and retained only those available for more than 95% of transit agencies to reduce potential bias from missing data. This resulted in 167 variables that reflect the operational and financial aspects of transit agencies. These variables were used for the kPCA and GMM analysis. Variables with higher missing data points, such as diesel consumption (missing for approximately one-third of agencies), were excluded from the clustering to avoid distortions due to data availability differences. A complete list of the 167 variables, along with their descriptions and mode-specific groupings, is provided in the public GitHub repository associated with this study (https://github.com/narslab/transit-typology-emissions.git). We considered the variables in two categories, namely, global (table 1) and mode-specific that apply to specific vehicle types (table 2). A summary of operational characteristics is shown in SI figure 1.

2.2. Dimensionality reduction

To reduce the dataset's dimensionality, we applied kPCA with a radial basis function (RBF) kernel. kPCA extends the traditional principal component analysis (PCA) by using kernel methods to capture nonlinear relationships, making it effective for datasets with complex patterns and interactions [39]. The RBF kernel models the data in a high-dimensional feature space, uncovering structures that linear PCA might miss [40]. We retained eight kernel principal components, explaining over 80% of the total variance (SI figure 2). This balance preserved most of the original information while simplifying the dataset for clustering and analysis. We used SHapley Additive exPlanations (SHAP) to interpret each kernel principal component [41]. For each component, we trained an XGBoost model with the original scaled features as predictors and the component values as the target. This quantified the contribution of each variable, highlighting key drivers such as total operating expenses, fleet size, and total vehicle miles for kPC1 (system size). Components were named according to their dominant variables (table 3). The top five variables for each component are shown in figure 2.

To evaluate the effectiveness of kernel principal component regression (kPCR) in predicting transit system characteristics, we trained an XGBoost model for each kPC. The scaled dataset was used to fit the model, and predictions were generated for each kPC. We evaluated the performance using mean squared

Table 1. Global variables used in the analysis. These variables apply to the entire system.

Variable
Agency total vehicles
Total vehicle miles (miles)
Vehicle operating EXP($)
Vehicle maintenance EXP($)
Non-vehicle maintenance EXP ($)
General administrative EXP ($)
Total operating EXP ($)
Total capital EXP ($)
Maximum service vehicles
Urban area (sq. miles)
Urban population
Service days operated
Emergency service days not operated
Non-statutory mixed traffic (miles)
Directional route miles (miles)

Table 2. Mode-specific variables used in the analysis. These variables are reported for each transit mode. *X* indicates the presence of a variable for a particular mode.

Variable	Aerial tramway	Commuter bus	Cable car	Commuter rail	Demand response	Demand taxi	Ferryboat	Heavy rail	Inclined plane	Jitney	Light rail	Bus	Monorail	Bus rapid transit	Streetcar	Trolleybus	Vanpool	Hybrid rail	Alaska railroad	Publico
Agency vehicles	X	X	X	X	X		X	X	X		X	X	X			X	X	X	X	X
Fare revenues($)	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X
Directional route miles (miles)	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X	X
Vehicle revenue miles (miles)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Vehicle revenue hours (Hours)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Unlinked passenger trips	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Passenger miles traveled (Miles)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Table 3. Kernel principal component (kPC) descriptions.

No.	Name	Description
kPC1	System Size	Represents the overall scale of the transit system, driven by large operating budgets, extensive vehicle fleets, high vehicle miles, and large service areas.
kPC2	Travel Demand	Captures the underlying size and population of the urbanized area, reflecting broader demand for mobility services.
kPC3	Small Demand Response	Reflects the volume of activity specifically in demand-response (DR) operations, including DR passenger miles, DR revenue hours, and DR passenger trip counts. Agencies with concentrated small-scale DR activity score high on this component.
kPC4	Flexible Demand	Represents demand for flexible, non-fixed-route services (DR and commuter services together). High demand-response vehicle miles, trips, and service intensity drive the model output.
kPC5	Fixed+Flexible Demand	Reflects hybrid systems characterized by the combined influence of fixed-route commuter bus services and demand-response operations, reflecting an integrated service structure rather than dominance by a single mode.
kPC6	Vanpooling	Captures vanpool intensity and associated support/administrative costs. Strong loadings come from vanpool vehicle counts, service activity, overhead expenses, and maintenance needs tied to vanpool operations.
kPC7	Sparse Service	Represents systems characterized by low service intensity and geographically dispersed operations. Key contributors include fewer service days, smaller urban populations, and a reliance on demand-response services to provide coverage in low-density areas.
kPC8	Demand Response Taxi	Captures fare-generating activity in DR and taxi-like services, including revenue hours, revenue miles, and passenger trips associated with DR-taxi operations.

error (MSE), mean absolute error (MAE), and the coefficient of determination (R^2). The results indicate that most components demonstrate high predictive accuracy, with R^2 values exceeding 0.95 for the majority of the kPCs. Table 4 summarizes the model performance for each kPC.

2.3. Clustering

We applied a GMM to group transit agencies into distinct clusters, using the extracted kernel principal components as features. To determine the optimal number of clusters, we performed a grid search over various cluster numbers, utilizing the full covariance type. Each number of clusters was evaluated using several criteria, including the Bayesian information criterion (BIC), Akaike information criterion (AIC), Negative log-likelihood, and the silhouette score. As shown in SI figure 3, these metrics indicate diminishing improvements beyond five clusters, with five representing a local optimum or elbow point across BIC, AIC, and silhouette score. We therefore selected five clusters as a balanced tradeoff between model optimality and interpretability.

2.4. Diesel consumption prediction

We developed a predictive model to estimate annual diesel consumption for transit systems lacking observed data. The model uses an XGBoost algorithm trained on detailed operational and financial indicators from the NTD, along with each system's assigned transit type from the GMM clustering. Including both original variables and cluster labels enables the model to capture system-level characteristics as well as typology-based similarities. Before finalizing the predictive framework, we compared four alternative model configurations, including: (1) using the original operational and financial variables with cluster assignments; (2) using the same variables without clusters; (3) using kPCs with clusters; and (4) using kPCs without clusters. We tuned each model using grid search with 5-fold cross-validation. As summarized in the SI table 1, the model using variables with cluster assignments achieved the lowest mean absolute percentage error (MAPE). We optimized the selected model using grid search with 5-fold cross-validation (SI table 2), achieving a MAPE of 24% and a median absolute percentage error of 19%. Figure 3 compares predicted and observed diesel consumption on the test set.

To assess whether model performance varies across transit system types, we computed test set error metrics separately for the transit types with observed diesel consumption (table 5). The model performs consistently across three of the four types (MAPE = 5%–39%). Because typology is only one of 167 predictors, and the model relies primarily on

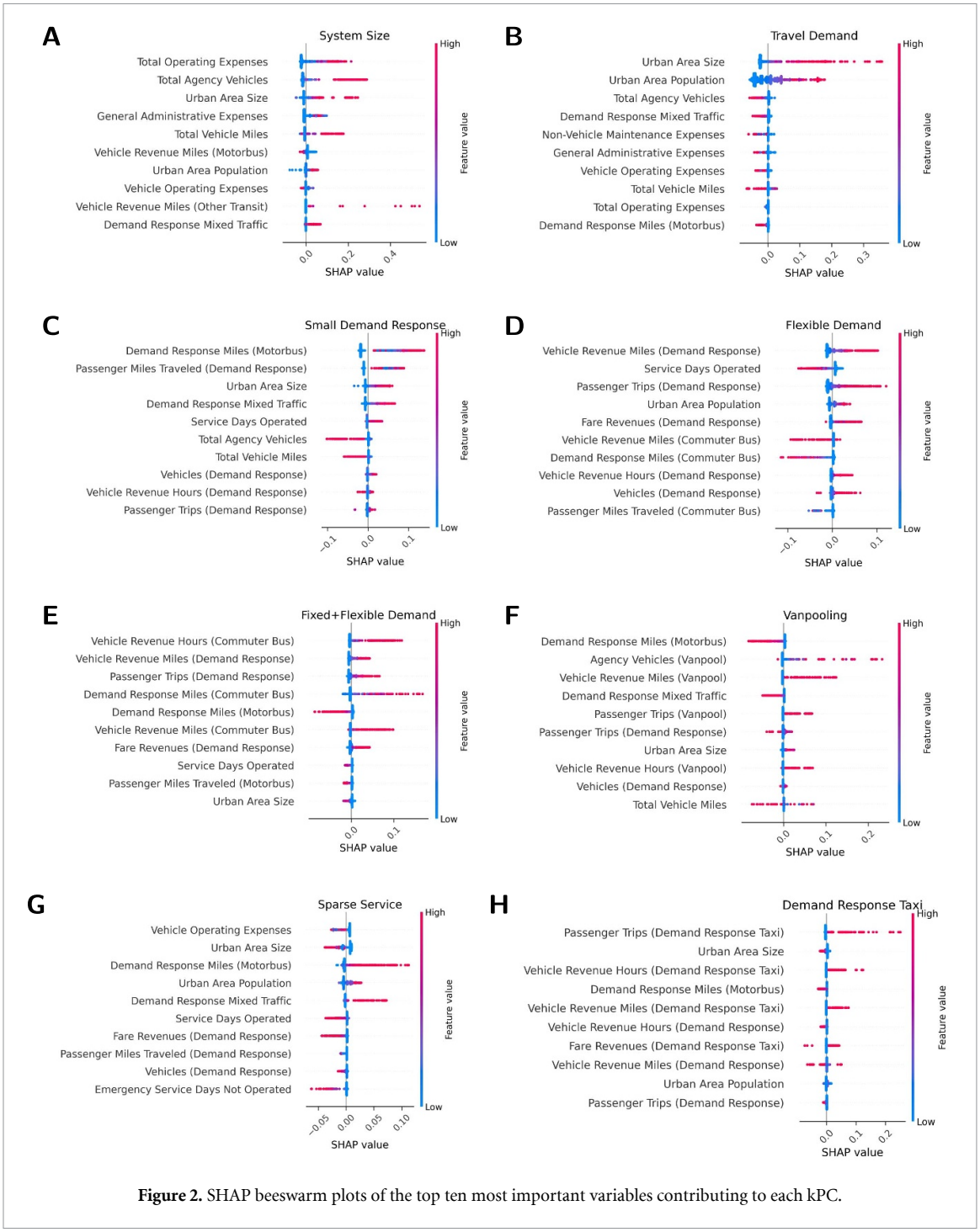


Figure 2. SHAP beeswarm plots of the top ten most important variables contributing to each kPC.

Table 4. Performance metrics for models predicting each kernel principal component from the indicator variables.

kPC	MSE (gal^2)	MAE (gal)	R^2
System size	0.001 939	0.021 209	0.960
Travel demand	0.000 137	0.005 997	0.989
Small demand response	0.000 621	0.012 221	0.946
Flexible demand	0.000 101	0.005 973	0.978
Fixed+flexible demand	0.000 087	0.005 373	0.977
Vanpooling	0.000 045	0.003 428	0.985
Sparse service	0.000 139	0.006 933	0.943
Demand response taxi	0.000 030	0.003 299	0.984

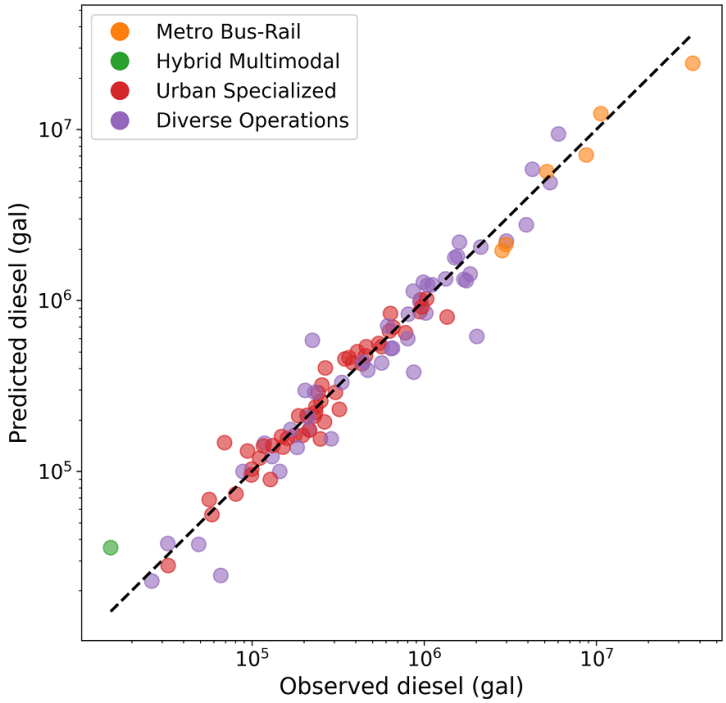


Figure 3. Predicted versus observed diesel (gal) for the test set.

Table 5. Cluster-specific model performance based on test-set predictions for transit system types with observed diesel data.

Cluster name	MSE (gal^2)	MAPE %)
Diverse operations	1.07×10^{11}	31
Hybrid multimodal	1.11×10^{8}	5
Metro Bus–Rail	1.86×10^{12}	39
Urban specialized	9.19×10^{9}	18

operational, fleet, and financial variables, extrapolating diesel consumption to the *Mid-Sized Urban* type remains reasonable.

To quantify predictive uncertainty, we adopted a transformed-target approach in which the XGBoost model is trained to predict

$$\log(1+y),$$

where y denotes annual diesel consumption. This transformation stabilizes the variance of residuals, mitigates heteroskedasticity, and ensures positive predictions after back-transformation. We computed the standard deviation of the log-space residuals using the held-out test set σ_ε, which represents the predictive dispersion of the model. For each transit agency, a 95% prediction interval (PI) is obtained as

$$\hat{y}_{\mathrm{PI}} = \exp(\hat{\mu} \pm 1.96\sigma_\varepsilon) - 1,$$

where $\hat{\mu}$ is the predicted log value. These intervals quantify uncertainty for individual transit agencies. We additionally reported σ_ε as a global summary measure of predictive uncertainty at the national and cluster levels. The system-level PIs are propagated through the well-to-tank (WTT) and well-to-wheel (WTW) emissions calculations to produce uncertainty bounds for emissions estimates.

To better understand the key drivers of diesel consumption in the model, we computed SHAP values for each feature. Figure 4 shows the SHAP values for the most important variables. Total operating exp, total vehicle miles, vehicle operating exp, vehicle maintenance exp, and urban population are the most influential features, indicating that systems with higher overall costs and demand tend to consume more diesel.

2.5. Emissions estimation

We estimated GHG emissions for diesel and electric bus operations using a WTW framework, which includes both direct tailpipe (or tank-to-wheel, TTW) and upstream fuel-cycle (or WTT) emissions. For diesel buses, we used a TTW emission factor for CO_2 of 10.21 kg CO_2 per gallon [42]. In addition to CO_2, diesel vehicles emit methane (CH_4) and nitrous oxide (N_2O) at rates of 0.0095 g and 0.0431 g per vehicle-mile, respectively. These were converted to kg per gallon using an average fuel economy of 3.40

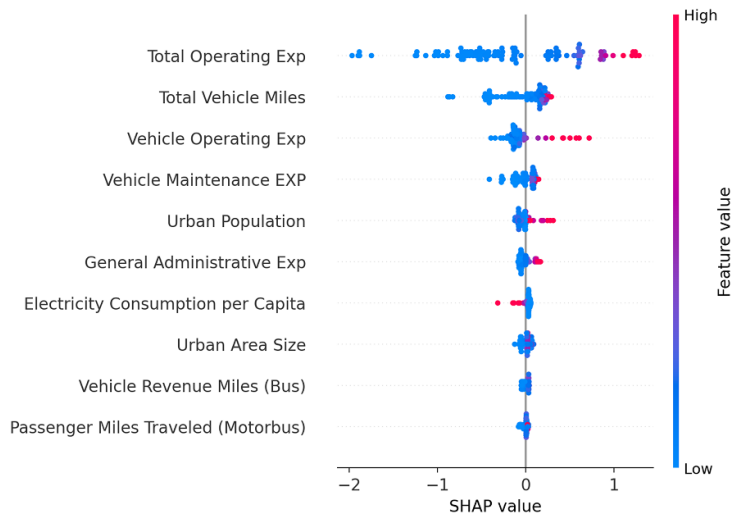


Figure 4. SHAP values indicating the contribution of key features to diesel consumption predictions.

miles per gallon (mpg) for diesel transit buses [43]. We applied global warming potentials from the IPCC Fifth Assessment Report: 28 for CH_4 and 265 for N_2O [44]. Summing all GHGs, the total TTW emission factor was 10.25 kg CO_2e per gallon. To estimate lifecycle emissions, we used the WTT emissions factor of 535.5 g CO_2e per liter of diesel (2.03 kg gallon^{-1}) [45]. The resulting WTW diesel emission factor is 12.28 kg CO_2e per gallon. A summary of all emission factors and assumptions is provided in SI table 3.

For electric operations, we estimated the electricity required to replace diesel consumption using the following conversion:

$$E = \frac{D}{\eta} \times \epsilon \tag{1}$$

where E is the electricity demand in kilowatt-hours (kWh), D is diesel consumption in gallons, η is the fuel economy of diesel buses (3.40 miles gallon^{-1}), and ϵ is the average energy use of electric buses (2.02 kWh mile^{-1}). We assigned each transit system to its emissions and generation resource integrated database (eGRID) subregion and applied subregion-specific WTT electricity emission factors ϕ_r from the U.S. Environmental Protection Agency [42]. Mapping of eGRID subregions to states and CO_2 emission factors is provided in SI table 4. The WTT electricity emissions were calculated as:

$$G^{\mathrm{WTT}} = \phi_r E \tag{2}$$

where G^{WTT} is the WTT emissions from electricity usage in kgCO_2e, E is the total electricity demand in megawatt-hours (MWh), and ϕ_r is the subregion-specific WTT emission factor (in kgCO_2e MWh^{-1}).

3. Results

3.1. Five transit system types uncovered

We identified five distinct transit system types using a GMM clustering, based on eight kernel principal components. These types reveal varying profiles of transit systems, characterized by differences in operational scales, geographic areas, and service levels. Figure 5 shows the distribution of kernel principal component scores across the five clusters and highlights the distinct operational and contextual profiles of each transit type. Table 6 summarizes the key indicators averaged by type, highlighting the operational and geographic characteristics of each transit type. The description of kPCs is provided in table 3.

To further explore the geographical distribution of the types, we visualize the transit types on a map of the U.S. (figure 6), which reveals significant geographical trends. For instance, transit types with higher operational scales and funding sources like *Metro Bus–Rail* systems, such as MTA New York City Transit (figure 6(A)), are concentrated in densely populated regions. *Mid-Sized Urban* systems, such as Dallas McKinney Avenue Transit Authority (figure 6(B)), are geographically dispersed and serve moderate urban areas with smaller operational scales. *Hybrid Multimodal* systems, such as South Jersey Transportation Authority (figure 6(C)), provide connectivity between urban centers and surrounding rural regions. In contrast, types characterized by specialized operations, like *Urban Specialized* systems such as Atlantic City transit (figure 6(D)), and smaller-scale operations with limited funding, like *Diverse Operations* systems such as Worcester Regional Transit Authority (figure 6(E)), are more dispersed across medium-sized cities and rural areas, respectively. For more details on the closest transit systems to the centroids of each cluster, and sample

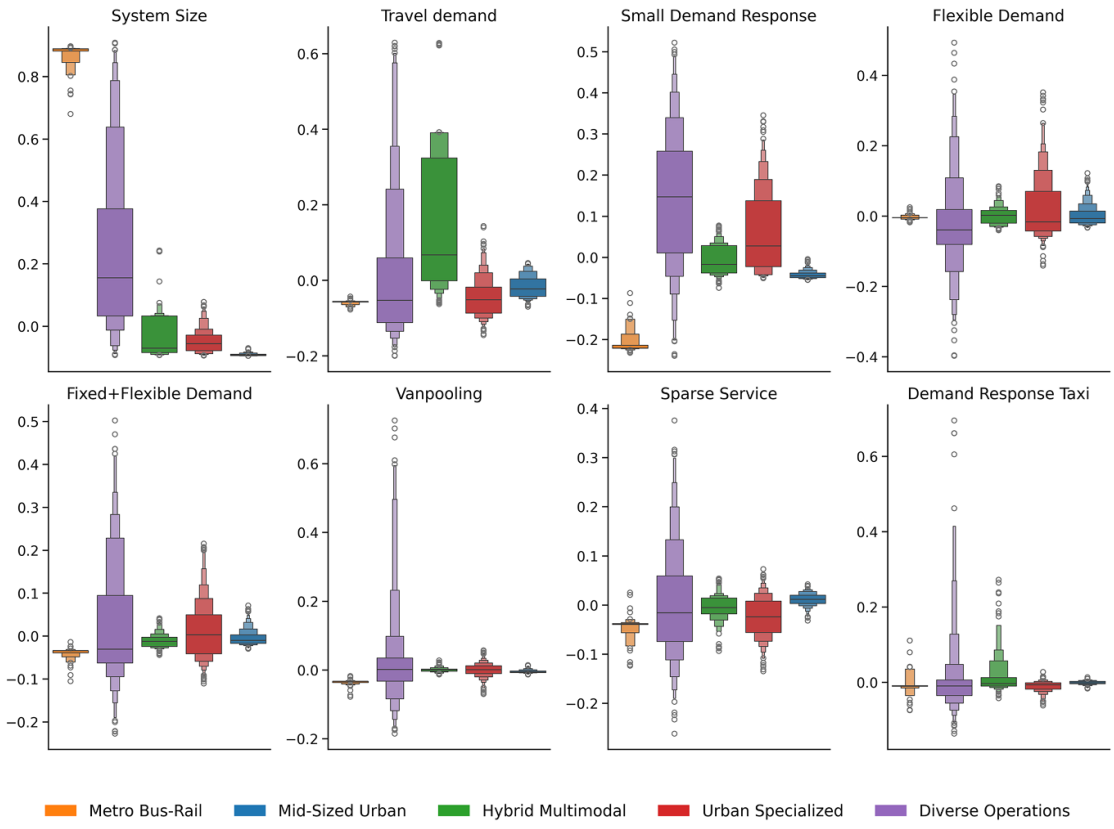


Figure 5. Distribution of kPC by transit system type.

Table 6. Key characteristics and summary statistics by transit type. Vehicle Miles and Vehicle Operating Expenses are not reported in the NTD for the *Mid-Sized Urban* type.

Indicator	Metro Bus–Rail	Mid-Sized Urban	Hybrid Multimodal	Urban Specialized	Diverse Operations
Urban Area (sq mi)	70 250	14 654	213 591	52 612	255 376
Population	171 732 121	31 377 970	24 667 545	60 617 411	277 414 792
Agency Vehicles	270 892	17 840	3837	29 103	125 786
Vehicle Miles (M miles)	2378	—	0.06	307	1299
Vehicle Operating EXP($M)	17 178	—	0.82	1350	6499

transit agencies grouped by city, see SI tables 5 and 6.

***Metro Bus–Rail*:** This type includes large urban transit systems, representing 4% (72 systems) of all U.S. transit systems and serving large metropolitan areas with an average service area population of approximately 2.38 million and an average primary urbanized area (UZA) spanning 976 square miles (sq. mi.). These systems exhibit the largest operational scale, with an average of 33 million vehicle miles annually, driven by diverse multimodal networks including buses, commuter rails, and subways. Concentrated in densely populated regions, particularly coastal cities and major urban hubs, notable examples include the MTA Bus Company, New York, NY, and the Massachusetts Bay Transportation Authority, Boston, MA.

***Mid-Sized Urban*:** This type represents 53% (1177 systems) of all U.S. transit systems, serving moderate urban areas with an average population of 27 000 and an average UZA of 12 sq. mi. While it is the most common type, these systems are relatively small in scale and lack reported data for many operational variables. Their services mainly include bus-based transit, vanpools, and demand-responsive options tailored to local needs. Geographically dispersed, they lack regional dominance. Notable examples include the RTS Genesee, Rochester, NY, and the Pacific Transit, Raymond, WA. In addition, this category is characterized by the notable absence of reported energy data in



Figure 6. Spatial distribution of transit system types. Each map highlights 20 representative cities to illustrate typical systems within each transit type.

the NTD, underscoring the need for further investigation into their energy usage and potential pathways for decarbonization.

***Hybrid Multimodal*:** This type represents 12% (253 systems) and operates in mixed urban-rural areas, with a service area population averaging 97 000 and an average UZA of 844 sq. mi. These systems balance moderate operational scales with an average of 15 million vehicle miles annually and emphasize demand-responsive services to bridge the connectivity gap between urban centers and surrounding rural regions. Notable examples include the Center for Pan

Table 7. Cluster-level contributions to TTW, WTT, and WTW emissions (percent of national total).

Cluster name	WTT (Mtone)	Contribution WTT(%)	TTW (Mtone)	Contribution TTW(%)	WTW (Mtone)	Contribution WTW(%)
Diverse operations	1.33	41.3	2.86	40.57	4.19	40.80
Hybrid multimodal	0.01	0.31	0.01	0.14	0.02	0.19
Metro Bus–Rail	1.56	48.45	3.55	50.35	5.11	49.76
Mid-sized urban	0.02	0.62	0.04	0.57	0.06	0.58
Urban specialized	0.30	9.32	0.59	8.37	0.89	8.67

Asian Community Services, Atlanta, GA, and the South Jersey Transportation Authority, Blackwood, NJ.

***Urban Specialized*:** Comprising 18% (400 systems) of all U.S. transit systems, this type operates in medium-sized urban areas, characterized by an average population of approximately 151 000 and an average UZA of 131 square miles. These systems demonstrate a balanced operational scale, averaging 4.28 million vehicle miles annually, and emphasize flexibility to urban dynamics, which is evident in their ability to serve diverse transit needs, including commuter operations, demand-responsive services, and specialized routes for events or tourism. Notable examples include the City of Los Angeles Department of Transportation, Los Angeles, CA, and the Beaver County Transit Authority, Rochester, PA, and City of Albany Transit System, Albany, OR.

***Diverse Operations*:** This type includes 13% (303 systems) and serves smaller, often less-densely populated regions with an average population of approximately 915 000 and an average UZA of 843 sq. mi. Despite their smaller scale, they provide a wide range of services, including buses, vanpools, and demand-responsive transit, to meet diverse regional needs. These systems operate an average of 4285 000 vehicle miles annually, reflecting their focus on tailored operations within limited geographic and population constraints. Notable examples are the Worcester Regional Transit Authority, Worcester, MA and the Southern California Regional Rail Authority, Los Angeles, CA.

3.2. Predictive model bridges transit system energy data gap

Actual diesel consumption data is available for only 531 transit systems (24%), spanning four distinct types. However, diesel consumption data is missing for 1674 systems (76%), including all transit systems in the *Mid-Sized Urban* type (SI figure 4). This substantial gap underscores the need for a robust estimation approach to support decarbonization planning. The results from the trained XGBoost model reveal that the total predicted diesel consumption for transit systems is approximately 689 million gallons nationwide. Scatter plots of the relationship between predicted diesel consumption and key variables are provided in SI figure 5. In addition to addressing the data gaps in reported consumption, the model allows us to identify key features associated with diesel use and examine whether energy patterns vary across transit types. This model provides a comprehensive understanding of transit sector diesel use and enables more accurate national-level accounting for decarbonization planning.

As shown in figures 7(A) and (B), the *Metro Bus–Rail* transit systems exhibit the highest values for both total diesel consumption and diesel consumption per capita, reflecting their large fleets, extensive service areas, and higher passenger demand. The *Urban Specialized* transit systems rank second in both measures, consistent with their high service intensity within focused urban areas. In contrast, *Hybrid Multimodal* and *Mid-Sized Urban* types show much lower diesel usage, showing smaller fleets, lower service intensity, or a greater reliance on non-diesel modes.

Table 7 reports the percentage contribution of each transit cluster to national TTW, WTT, and total WTW diesel-based emissions. TTW emissions account for the majority of total WTW emissions, while variation in WTT emission factors across eGRID subregions leads to relatively small shifts in cluster-level shares.

3.3. Typology-based energy estimates show uneven potential for emissions reduction

Using predicted diesel consumption, we estimate that U.S. transit systems emit approximately 7.05 $MMTCO_2e$ annually from direct fuel combustion TTW. To provide a full lifecycle perspective, we also incorporate upstream emissions associated with diesel production and distribution WTT, adding 3.22 $MMTCO_2e$. Combined, the total WTW emissions for all 2205 transit systems are estimated at nearly 10.27 $MMTCO_2e$ per year.

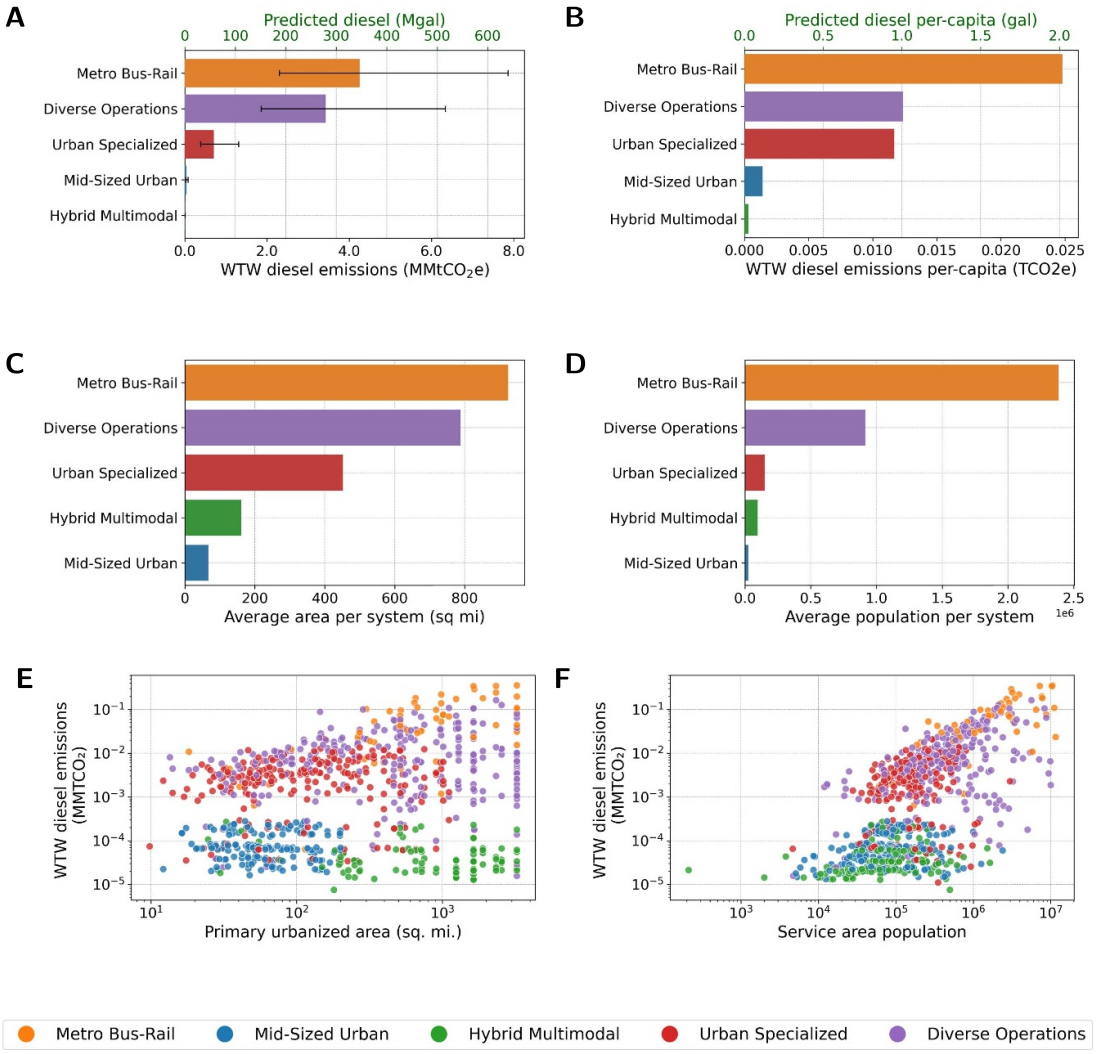


Figure 7. Predicted emissions across 2205 U.S. transit systems, grouped by transit type: (A) predicted annual diesel use (Mgal) and WTW emissions ($MMTCO_2e$) with 95% prediction intervals, (B) per-capita WTW diesel emissions (TCO_2), (C) average service area (sq mi) per system, and (D) average population served per system. (E) WTW emissions versus primary urbanized area (sq. mi.), (F) WTW emissions versus service area population.

We summarize the predicted total and per-capita diesel use and WTW emissions across the five identified transit types (figures 7(A) and (B)). The *Metro Bus–Rail* type dominates national emissions, contributing about 50% of the total despite representing only 4% of all 2205 systems analyzed. These systems operate with larger fleet sizes and serve the largest population centers. In contrast, types such as *Hybrid Multimodal* and *Mid-Sized Urban* contribute the least, reflecting smaller fleet sizes and more localized service areas. Figures 7(C) and (D) provide further insight into the spatial and demographic reach of transit systems by type. *Metro Bus–Rail systems cover relatively larger areas per system and serve larger populations.* In contrast, *Diverse Operations* and *Hybrid Multimodal* systems operate over wide areas per system but serve fewer people, reflecting the more dispersed and rural nature of these networks. These structural differences help explain observed variations in fuel consumption and emissions across clusters. Figures 7(E) and (F) illustrate how WTW emissions scale with the primary UZA and the population, respectively. The upward trends reflect the impact of system size and activity level on overall emissions.

Metro Bus–Rail systems produce the highest WTW emissions due to their large-scale operations, while *Hybrid Multimodal* systems have the lowest emissions, reflecting smaller service areas and lower energy needs. Currently, U.S. transit systems collectively emit approximately 10.27 $MMTCO_2e$ WTW annually, with 7.05 $MMTCO_2e$ coming from TTW emissions and the remaining 3.22 $MMTCO_2e$ from WTT emissions. In comparison, the total projected annual emissions after complete fleet electrification amount to 1.82 $MMTCO_2e$, representing a significant reduction of 82% in emissions. This highlights the substantial decarbonization potential of transitioning to electric transit fleets.

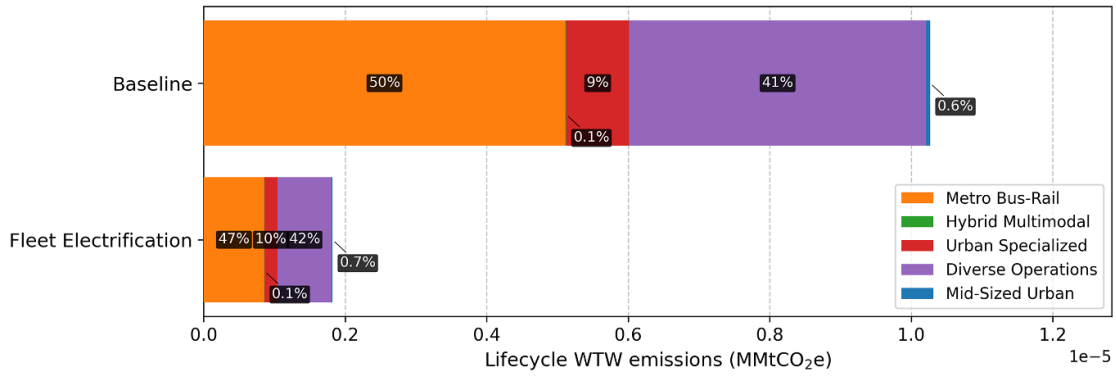


Figure 8. Annual WTW emissions under Baseline and Fleet Electrification scenarios.

To evaluate the impact of electrification, we consider two long-term decarbonization scenarios: *Fleet Electrification* and *Grid Decarbonization*. In the *Fleet Electrification* scenario, all diesel-powered transit fleets are replaced with battery-electric buses while the electricity mix reflects the current U.S. grid. Because tailpipe emissions TTW are eliminated entirely, the remaining emissions are attributed solely to the upstream WTT stage of electricity generation. Under this transition, total WTW emissions fall to 1.82 $MMTCO_2e$ annually—an 82% reduction from the diesel baseline of 10.27 $MMTCO_2e$.

In the *Grid Decarbonization* scenario, electrification is combined with a fully decarbonized electricity grid, in which all electricity is generated from renewable sources such as wind, solar, and hydro. This eliminates both TTW and WTT emissions, resulting in zero total lifecycle emissions. Relative to *Baseline* scenario, this yields an additional reduction of 1.82 $MMTCO_2e$, or 18 percentage points. Compared to the *Baseline* scenario, this represents a complete (100 %) reduction in lifecycle emissions. These results, summarized in figure 8, highlight the importance of pairing fleet electrification with grid decarbonization to achieve the greatest emissions reduction.

4. Conclusion

We developed a data-driven typology for 2205 U.S. transit agencies and reduced 167 indicators to 8 kernel principal components that capture fundamental differences in operating scale, service patterns, geography, and finances. Using these components, we identified five system types-*Metro Bus–Rail*, *Mid-Sized Urban*, *Hybrid Multimodal*, *Urban Specialized*, and *Diverse Operations*, which characterize key operational patterns and explain much of the variation in diesel use and emissions across transit agencies.

We addressed a major national data gap by training an XGBoost model to estimate diesel consumption for systems with missing energy data (76% of agencies). The model yields a national baseline (2022) of 688 million gallons of diesel and 10.27 $MMTCO_2e$ WTW emissions annually, with *Metro Bus–Rail* systems accounting for the largest share of absolute emissions. We then quantified the decarbonization potential under two scenarios. Under *Fleet Electrification* (battery-electric buses on today's grid), total WTW emissions fall to 1.82 $MMTCO_2e$, an 82% reduction relative to the 10.27 $MMTCO_2e$ diesel baseline, driven primarily by the elimination of TTW emissions. Under *Grid Decarbonization*, these remaining 1.82 $MMTCO_2e$ emissions are eliminated, resulting in zero lifecycle emissions.

This integrated typology-prediction framework enables benchmarking within and between types and supports transfer of information from data-rich to underreported agencies. Agencies in the *Metro Bus–Rail* type offer the largest absolute reduction potential and should be prioritized for rapid bus electrification and charging deployment. *Mid-Sized Urban* and *Hybrid Multimodal* systems can leverage the typology to stage electrification where it is most impactful and to improve data reporting for continuous tracking. Across all types, pairing fleet conversion with programs that accelerate grid decarbonization maximizes long-run climate benefits.

Future work can extend this framework, evaluating additional operational strategies such as scheduling and charging optimization by transit system type. By combining a national typology with predictive energy modeling and clear scenario analysis, this study provides a practical foundation for targeting investments, closing data gaps, and accelerating transit decarbonization in the U.S.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://github.com/narslab/transit-typology-emissions.git.

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Author contributions

Conceptualization: M A and J O; methodology: M A and J O; investigation: M A and J O; coding: M A; writing–original draft: M A; writing–review & editing: M A and J O; resources: M A and J O; supervision: J O.

Conflict of interest

The authors declare no conflict of interest.

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